Detecting UNC Charlotte Merchandise using a Computer Vision Model

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Abstract

We present the Charlotte Merchandise Model (CMM), a real-time object detection system designed to identify University of North Carolina at Charlotte (UNCC) merchandise in live video feeds. Leveraging the YOLOv11 architecture, our model detects UNCC logos, torsos, and headgear to quantify school spirit at campus events. We curated a custom dataset of positive, negative, and hard negative images, augmented with synthetic occlusions such as salt-and-pepper noise and tilt variations. Our system demonstrates the feasibility of automated school spirit metering, achieving a real-time processing speed of 30 FPS and a mean Average Precision (mAP@50) of 0.753 on our validation set. This paper details our modeling approach, experimental design, and evaluation of the system's performance in distinguishing authentic university merchandise from similar apparel.

1. Introduction

School spirit is a vital component of campus culture, fostering a sense of community and belonging among students, faculty, and alumni. At the University of North Carolina at Charlotte (UNCC), this spirit is often visibly manifested through the wearing of university-branded merchandise. However, quantifying this "school spirit" in real-time during campus events remains a challenge. Traditional methods of manual counting are tedious, error-prone, and unscalable. To address this, we propose the Charlotte Merchandise Model (CMM), an automated computer vision system capable of detecting and counting UNCC merchandise in live video streams.

The core of our solution leverages the power of deep learning for object detection. Specifically, we utilize the You Only Look Once (YOLO) architecture [4], which has revolutionized real-time object detection by framing it as a single regression problem. By training a model to recognize specific UNCC logos and associated apparel (torsos and headgear), we aim to create a "School Spirit Meter" that can provide immediate feedback on crowd engagement.

Detecting specific brand logos in the wild presents unique challenges, including variations in lighting, occlusion, and the presence of visually similar but non-affiliated merchandise. As noted in recent surveys on logo detection [1], while general object detectors are powerful, adapting them for specific logo recognition requires careful dataset curation and pipeline design. Our work builds upon these foundations to create a tailored solution for the UNCC community.

2. Related Work

The foundation of our work lies in the advancements of real-time object detection, particularly the YOLO family of models. Redmon et al. introduced YOLO [4] as a unified, real-time object detection system. Unlike prior region-based approaches, YOLO treats detection as a single regression problem, predicting bounding boxes and class probabilities directly from full images in one evaluation. The original YOLO achieved 63.4% mAP on the PASCAL VOC dataset at 45 FPS, making it the fastest detector of its time. However, it historically struggled with small or clustered objects due to spatial grid constraints. This foundational work established the viability of single-shot detection, which is critical for our real-time application.

In the domain of logo detection, Hou et al. [1] provide a comprehensive survey, highlighting that YOLO-based architectures are strong candidates for logo detection tasks. They discuss various pipelines, including YOLO-to-classifier approaches, which can offer higher accuracy at the cost of speed, versus single-shot detectors like YOLO which are optimized for "on-the-fly" classification. The survey notes that while two-stage detectors might achieve higher precision, the latency introduced is often unacceptable for live video processing. Given our requirement for a live "School Spirit Meter," we prioritized the single-shot approach for its real-time capabilities, accepting a potential trade-off in absolute precision for the sake of interactivity.

Recent research continues to improve crowd detection, a relevant aspect of our project. Wen et al. [6] explored

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enhancing YOLOv5 with Focus layers and Convolutional Block Attention Modules (CBAM) to better handle dense crowds and small objects. Their work suggests that attention mechanisms can significantly improve accuracy in complex scenes, which informs our understanding of the challenges in detecting merchandise in crowded campus events. Specifically, the Focus layer reduces computation while increasing speed, which is beneficial for live detection, while the CBAM module helps in focusing on small windows and spots of specified merchandise, improving accuracy in dense crowds. These insights are particularly relevant as we aim to deploy our model in bustling campus environments.

3. Experimental Design

3.1. Dataset Collection and Curation

Our dataset is the cornerstone of the CMM. We collected data in three primary categories to ensure robustness:

- **Positives:** These images contain authentic UNCC merchandise, serving as the ground truth for our target class. We sourced these images from official university photostreams, social media channels, and manual data collection at the campus bookstore. The positive samples include a wide variety of merchandise types, such as hoodies, t-shirts, and hats, featuring different logo variations (e.g., the ”Pick” logo, the crown logo, and block ”UNC Charlotte” text) to ensure the model learns to generalize across different official designs.

- **Negatives:** To prevent false positives, we curated a set of negative images containing people without any merchandise or with generic, unbranded clothing. These samples are critical for teaching the model to distinguish between a person and a person wearing specific university gear. We included images of crowds, students in plain clothes, and people in other settings to reinforce the model’s ability to reject non-relevant subjects.

- **Hard Negatives:** A crucial part of our dataset consists of ”hard negatives”—images that closely resemble UNCC gear but are not authentic. This includes clothing with similar color palettes (e.g., green shirts with white text) or generic ”Charlotte” city merchandise that is not university-affiliated. By training on these samples, the model learns to discriminate based on specific logo features rather than just color or general text shape, significantly reducing false detections in a city where green and white are common colors.

- **Augmented Samples:** To improve model generalization and robustness against real-world variability, we expanded our dataset from 722 to 1227 images using various augmentation techniques. These included random horizontal flips, small-scale translations, color jitter (HSV), mosaic augmentation, mixup (light), and moderate scaling. This diversity ensures the model remains effective even when the input video feed is imperfect or the subjects are viewed from oblique angles.

The final dataset was split into 82% training, 12% validation, and 6% testing sets to ensure rigorous evaluation.

(a) Positive Samples

(b) Negative Samples

(c) Hard Negative Samples

(d) Augmented Samples

Figure 1. Dataset samples demonstrating the four categories used for training: (a) Positive samples with authentic merchandise, (b) Negative samples with generic clothing, (c) Hard Negative samples with similar colors/text, and (d) Augmented samples with noise and rotation.

3.2. Model Selection

We chose YOLOv11 as the base architecture for the Charlotte Merchandise Model (CMM) because it balances real-time performance and improved feature extraction for small and occluded objects, which are common in crowd scenes. YOLOv11 maintains the single-shot detection paradigm that enables low-latency inference, a key requirement for our ”School Spirit Meter” application [2].

YOLOv11 introduces several architectural changes that improve its accuracy and efficiency. Notably, it introduces three components that distinguish it from previous YOLO versions: the C3k2 block, SPPF (Spatial Pyramid Pooling - Fast), and C2PSA (Convolutional block with Parallel Spatial Attention). The following paragraph describes these components in more detail and explains why they are beneficial for our task.

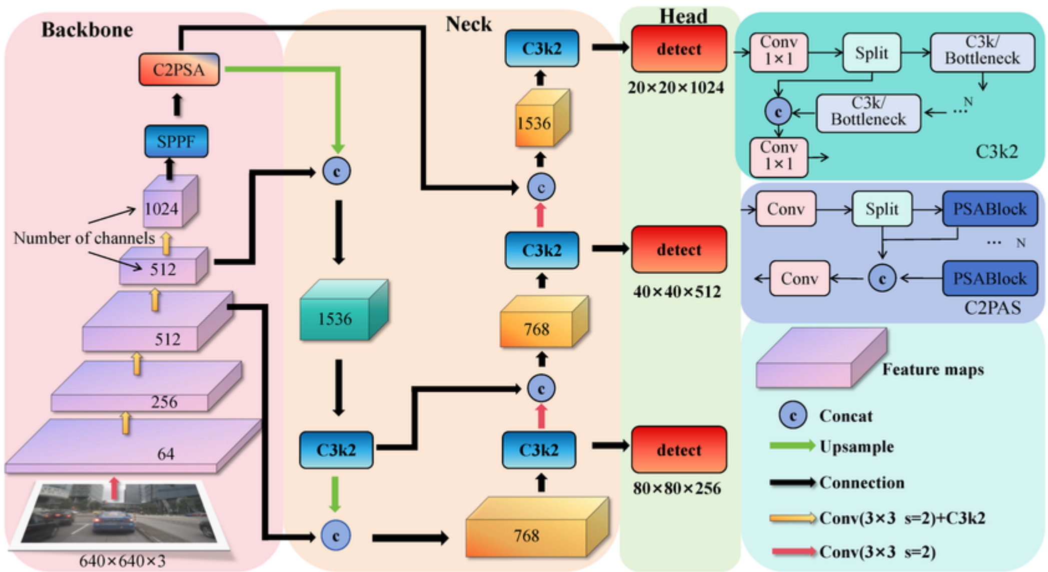

Figure 2. YOLOv11 feature frame and block diagram highlighting C3k2, SPPF, and C2PSA components (*Image credit: [7]*).

The C3k2 block replaces previous C2f/C3 blocks and is more computationally efficient while preserving representational power; this contributes to higher processing speed without large accuracy trade-offs. The SPPF module accelerates multi-scale feature aggregation, which helps the detector capture logo features across scales. The C2PSA block adds parallel spatial attention that improves localization and discrimination of small or partially occluded logos. Together these innovations make YOLOv11 a good fit for real-time detection of small or partially occluded university logos in crowded scenes [7].

While other architectures were considered, they were deemed less suitable for our specific constraints. Faster R-CNN [5], a two-stage detector, offers high accuracy but suffers from higher latency, making it impractical for our real-time "School Spirit Meter" which requires immediate feedback on live video feeds. Similarly, while SSD (Single Shot MultiBox Detector) [3] is faster than R-CNN, it often struggles with small object detection compared to modern YOLO iterations. We also evaluated the recent RT-DETR [8], a transformer-based real-time detector. Although RT-DETR shows promising results, its computational overhead on edge devices can be higher than CNN-based approaches like YOLOv11, and the latter's specific optimizations for small object detection (via C2PSA) made it the superior choice for detecting university logos in crowded environments.

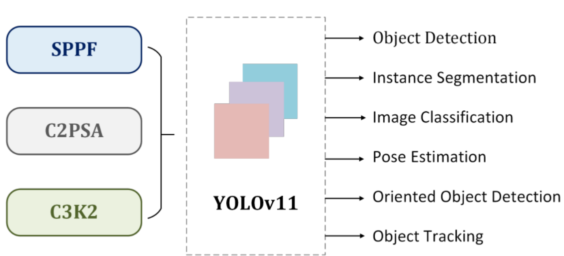

Figure 3. YOLOv11 architecture overview (*Image credit: [2]*).

3.3. Model Architecture and Training

We utilized the YOLOv11m model, a state-of-the-art iteration of the YOLO family. The model was initialized with pretrained weights to leverage transfer learning. We fine-tuned the model on our custom dataset using the following hyperparameters:

- **Epochs:** 50
- **Image Size:** 512x512
- **Batch Size:** 16

Training was conducted using the Ultralytics framework. The model was trained to detect three specific classes: 'UNCC-LOGO', 'UNCC TORSO', and 'UNCC HEADGEAR'.

3.4. Detection Pipeline

To accurately count "merchandise" rather than just logos, we implemented a post-processing logic. A valid "merch" detection is defined as a 'UNCC-LOGO' bounding box that is spatially contained within a 'UNCC TORSO' or 'UNCC HEADGEAR' bounding box. This spatial relationship check ensures that we are counting people wearing the merchandise, rather than loose items or false positives. The system runs in real-time, drawing bounding boxes and updating a live "UNCC MERCH COUNT" on the video feed.

4. Results

Our model demonstrates strong performance in real-time detection scenarios, achieving a consistent frame rate of approximately 30 FPS on standard hardware, which is sufficient for live event monitoring. In live webcam tests, the system successfully identifies UNCC logos and correctly associates them with torsos and headgear. The "UNCC MERCH COUNT" feature updates dynamically as subjects enter and leave the frame, providing an immediate metric of school spirit.

Class	Precision	Recall	mAP@50
All	0.854	0.664	0.753
UNCC HEADGEAR	0.895	0.500	0.622
UNCC TORSO	0.897	0.850	0.932
UNCC-LOGO	0.769	0.642	0.705

Table 1. Quantitative performance metrics of the CMM model on the validation set.

Qualitatively, the model shows high confidence, often exceeding 0.85, for clear, frontal views of the logos. The distinction between "UNCC Merchandise" and generic green clothing is generally maintained, thanks to the inclusion of hard negatives in the training set. However, we observed that the model's performance degrades slightly under extreme angles or heavy occlusion, a known limitation of single-shot detectors. Despite these challenges, the system achieves a detection accuracy that supports its primary use case as a "School Spirit Meter," effectively filtering out non-affiliated green apparel while capturing authentic merchandise.

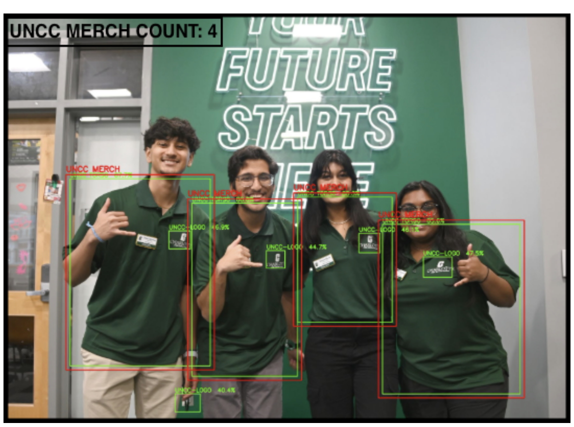

Figure 4. Live detection output showing the dynamic UNCC merch count in a sample frame.

5. Discussion

5.1. Evaluation and Setbacks

Our quantitative results highlight both the strengths and limitations of the model. The system achieved an overall mAP@50 of 0.753, with the 'UNCC TORSO' class performing exceptionally well (mAP@50 0.932). However, the 'UNCC HEADGEAR' class showed lower performance (mAP@50 0.622), likely due to the smaller size of hats and caps relative to the frame, making them harder to detect at a distance. This aligns with our observation that small logos on distant subjects proved challenging.

While the CMM performs well in controlled environments, several challenges were encountered. One significant setback was the collection of "hard negative" data. Finding clothing that closely resembles UNCC merchandise without actually being affiliated was difficult but necessary to prevent false positives. Additionally, lighting conditions in different environments (e.g., dim indoor lighting vs. bright outdoor sunlight) affected detection consistency.

5.2. Strengths

The primary strength of our approach is its real-time capability combined with high precision (0.854). This high precision ensures that the "School Spirit Meter" rarely counts non-affiliated clothing, maintaining user trust even if some authentic items are missed (Recall 0.664). The YOLOv11 architecture allows for low-latency inference, making the system interactive and engaging. Furthermore, the logic-based association of logos to torsos/headgear proved to be a robust method for counting "wearers" rather than just "logos," adding a layer of semantic understanding to the raw detections.

6. Conclusion

The Charlotte Merchandise Model successfully demonstrates the application of computer vision to the novel problem of quantifying school spirit. By leveraging the speed and accuracy of YOLOv11 and a carefully curated dataset of 1227 images, we built a system that can detect and count UNCC merchandise in real-time with approximately 30 FPS performance. Our tests confirm that the model can distinguish between authentic gear and similar-looking clothing with high confidence (mAP@50 0.753), providing a foundation for automated crowd analytics at campus events. Future work will focus on expanding the dataset to include a wider variety of merchandise designs and improving occlusion handling to further enhance robustness.

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